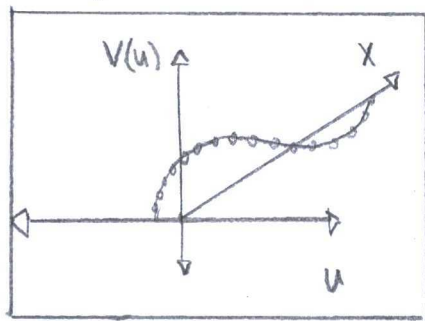


# Chapter 3: Path Integral:

## Exercise #1:

(pg 101)



"Path integral"

Position:  $y(x,t) = A \sin(kx - \omega t)$

Velocity:  $\dot{y}(x,t) = \omega A \cos(kx - \omega t)$

Kinetic Energy:

$$K.E. = \int \frac{1}{2} \mu (-\omega A \cos(kx - \omega t))^2 dx$$

$$= \frac{1}{2} \mu \omega^2 A^2 \int_0^L \cos^2(kx - \omega t) dx$$

... when  $\mu = \frac{\text{mass}}{\text{Length}}$

$$= \frac{1}{2} \mu \omega^2 A^2 \left( \frac{1}{2} L + \frac{\sin(2kL)}{4k} \right)$$

|              |
|--------------|
| Wavevector:  |
| $k = 2\pi/L$ |

$$= \frac{1}{4} \mu \omega^2 A^2$$

The continuum lattice model in Chapter 1 had an integral over a "path" from 0 to L, but with a Hamiltonian rather than a direct integral to Kinetic energy.

Limit ( $m \rightarrow 0$ ):

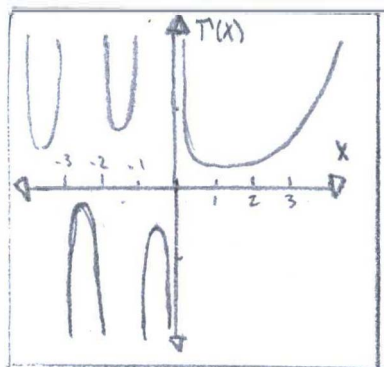
$$\lim_{m \rightarrow \infty} Z = \lim_{m \rightarrow \infty} \text{tr}(e^{-\beta H})$$

$$= \lim_{m \rightarrow \infty} \text{tr}(e^{-\beta(K.E. + V)})$$

$$= \lim_{m \rightarrow \infty} \text{tr}(e^{-\beta(\frac{1}{4}\mu\omega^2 A^2 + \hat{V})})$$

$$= 0$$

Exercise #2:  
(pg 105)



"Gamma  
Function"

(from problem) "Gamma Function"-modified

$$\Gamma(z+1) = \int_0^{\infty} dx \cdot x^z e^{-x}$$

Sterling Approximation:

$$\begin{aligned} \Gamma(z+1) &= \int_0^{\infty} dx \cdot x^z \cdot e^{-x} \\ &= \int_0^{\infty} dx \cdot e^{-x + s \ln x} \\ &\approx \int_0^{\infty} dx \cdot e^{-s + s \ln s - x^2/2s} \\ &= e^{s(\ln s - 1)} \cdot \sqrt{2\pi s} \end{aligned}$$

Minimum to Exponent:

$$F(x) = -x + s \ln x$$

$$F'(x) = -1 + \frac{s}{x}$$

$$= 0$$

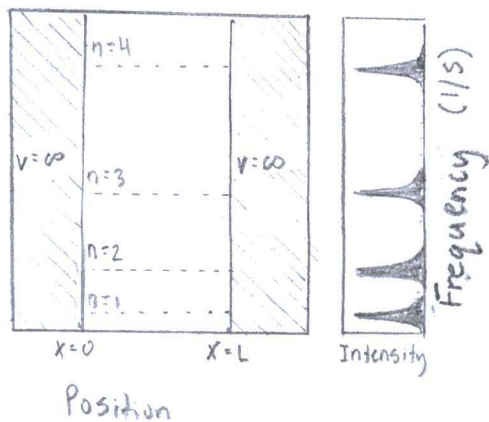
$$x_{\min} = s$$

Integral of  
a Gaussian  
Distribution

$$\int e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

# Exercise #3

(pg 107)



"Quantum particle  
in a well"

## (3.27) "Free Particle Operator"

$$G_{\text{free}}(q_F, q_i, t) = \langle q_F | e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} | q_i \rangle \Theta(t)$$

$$= \left( \frac{m}{2\pi i \hbar t} \right)^{1/2} e^{\frac{i}{\hbar} \frac{m (q_F - q_i)^2}{2t}} \Theta(t)$$

This integral

↑  
Heavyside \$\Theta\$-function

$$\langle q_F | e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} | q_i \rangle = \int_{-\infty}^{\infty} \frac{e^{-\frac{i}{\hbar} \frac{p^2 t}{2m}}}{\sqrt{2\pi}} \cdot e^{\frac{i}{\hbar} \frac{p (q_F - q_i)}{t}} \frac{e^{\frac{i}{\hbar} \frac{p (q_i - q_F)}{t}}}{\sqrt{2\pi}} dp$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \left[ \frac{p^2 t}{2m} + \frac{(q_F - q_i) p}{t} \right]} dp$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \left[ \frac{1}{2m} \left( p - \frac{m(q_F - q_i)}{t} \right)^2 - \frac{m(q_F - q_i)^2}{2t^2} \right] t} dp$$

$$= \frac{e^{-\frac{i}{\hbar} \frac{m(q_F - q_i)^2}{2t}}}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \left[ \frac{1}{2m} \left( p - \frac{m(q_F - q_i)}{t} \right)^2 \right] t} dp$$

$$= \frac{e^{-\frac{i}{\hbar} \frac{m(q_F - q_i)^2}{2t}}}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i p'^2 t}{2\hbar m}} dp'$$

... \$p' = p - \frac{m(q\_F - q\_i)}{t}\$

$$= \frac{e^{-\frac{i}{\hbar} \frac{m(q_F - q_i)^2}{2t}}}{2\pi} \left( \frac{2\pi}{it/\hbar m} \right)$$

$$= \left( \frac{m}{2\pi i \hbar t} \right)^{1/2} e^{-\frac{i}{\hbar} \frac{m(q_F - q_i)^2}{2t}}$$

(3.9) "Gaussian Integral"

$$\int_{-\infty}^{\infty} dx e^{-\frac{a}{2}x^2} = \sqrt{\frac{2\pi}{a}}$$

$$G_{\text{Free}}(q_F, q_i, t) [\text{units} = \hbar^{-1}] = \left( \frac{m}{2\pi i \hbar t} \right)^{1/2} e^{-\frac{i}{\hbar} \frac{m(q_F - q_i)^2}{2t}}$$

Exercise #4:  
(pg 107)

(3.1) "Time evolution operator"

$$U(t, t') = e^{-\frac{i}{\hbar} \hat{H}(t-t')} \cdot \Theta(t-t')$$

(3.7) "Action Integral"

$$\langle q_F | e^{-\frac{i}{\hbar} \hat{H}t} | q_i \rangle = \int_{\substack{q(t)=q_F \\ q(0)=q_i}} e^{-\frac{i}{\hbar} \int_0^t dt' V(q)} \cdot dq$$

$$\begin{aligned} \langle p' | U(t, t') | p \rangle &= \langle p' | e^{-\frac{i}{\hbar} \hat{H}t} | p \rangle \\ &= \langle p' | e^{-\frac{i}{\hbar} \int_0^t V(p) dt} | p \rangle \\ &= \langle p' | 1 - \frac{i}{\hbar} \int_{t_0}^t V(p) dt | p \rangle \end{aligned}$$

Taylor Expansion:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$= \langle p' | p \rangle - \frac{i}{\hbar} \langle p' | \int_{t_0}^t V(p) dt | p \rangle$$

$$= -\frac{i}{\hbar} \langle p' | \int_{-\infty}^{\infty} e^{-iH_0 t/\hbar} \cdot V \cdot e^{iH_0 t/\hbar} | p \rangle$$

(Orthogonal)

Inner products:

$$\begin{aligned} \langle p' | p \rangle &= \delta(p' - p) \\ &= \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases} \end{aligned}$$

$$= -\frac{i}{\hbar} \int_{-\infty}^{\infty} e^{-iE(p)t/\hbar} \cdot \langle p' | V | p \rangle \cdot e^{iE(p)t/\hbar} dt$$

$$= -\frac{i}{\hbar} \langle p' | V | p \rangle \int_{-\infty}^{\infty} e^{i(E(p') - E(p))t/\hbar} dt$$

$$\lim_{t \rightarrow \infty} \langle p' | U(t, t') | p \rangle = \lim_{t \rightarrow \infty} -\frac{i}{\hbar} \langle p' | V | p \rangle \int_{-\infty}^{\infty} e^{i(E(p') - E(p))t/\hbar} dt$$

$\propto \delta(E(p) - E(p'))$  ... say zero

$$\lim_{t' \rightarrow -\infty} \langle p' | U(t, t') | p \rangle = \lim_{t \rightarrow \infty} -\frac{i}{\hbar} \langle p' | V | p \rangle \int_{-\infty}^{\infty} e^{i(E(p') - E(p))t/\hbar} dt$$

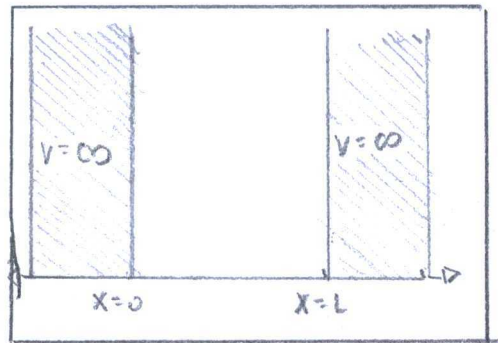
$\propto \delta(E(p) - E(p'))$  ... say one.

(From problem) "Born scattering amplitude"

$$\langle p' | U(t \rightarrow \infty, t' \rightarrow -\infty) | p \rangle = -\frac{i}{\hbar} e^{-i(t-t')E(p)/\hbar} \delta(E(p) - E(p')) \langle p' | V | p \rangle$$

Exercise #5 :

(pg 109)



"infinite  
square  
well"



# (3.27) "Free Particle Propagator"

$$\begin{aligned}
 G_{\text{free}}(q_F; q_i; t) &= \langle q_F | e^{-\frac{i}{\hbar} \hat{p}^2 \frac{t}{2m}} | q_i \rangle \Theta(t) \\
 &= \int_{-\infty}^{\infty} \frac{e^{-\frac{i}{\hbar} p q_F}}{\sqrt{2\pi\hbar}} \cdot e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} \cdot \frac{e^{-\frac{i}{\hbar} p q_i}}{\sqrt{2\pi\hbar}} dp \\
 &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} \cdot e^{-\frac{i}{\hbar} p(q_F - q_i)} dp
 \end{aligned}$$

As an Ensemble,

$$\begin{aligned}
 &\sum_{m=-\infty}^{\infty} [\langle q_F | e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} | q_i \rangle - \langle q_i | e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} | q_F \rangle] e^{-\frac{2ipmL}{\hbar}} dp \\
 &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} \left[ e^{\frac{i}{\hbar} p q_F} - e^{-\frac{i}{\hbar} p q_F} \right] e^{-\frac{i}{\hbar} p q_i} \cdot e^{-\frac{2ipmL}{\hbar}} dp \\
 &= \frac{i}{\pi\hbar} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} \cdot \sin\left(\frac{i}{\hbar} p q_F\right) \cdot e^{-\frac{i}{\hbar} p q_i} \sum_{m=-\infty}^{\infty} e^{-\frac{2ipmL}{\hbar}} dp \\
 &= \frac{i}{\pi\hbar} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} \sin(k_n q_F) \cdot e^{-\frac{i}{\hbar} p q_i} \cdot \sum_{m=-\infty}^{\infty} e^{-2\pi i n m} dp \\
 &= \frac{2i}{L} \sum_{n=-\infty}^{\infty} e^{-\frac{iEt}{\hbar}} \cdot \sin(k_n q_F) e^{\frac{i}{\hbar} p q_i} \\
 &\equiv \text{Re} \left( \frac{2}{L} \sum_{n=1}^{\infty} e^{-\frac{iEt}{\hbar}} \sin(k_n q_F) \sin(k_n q_i) \right)
 \end{aligned}$$

Identities in the lines above:

| <u>Momentum</u>             | <u>Energy</u>                  | <u>sine to exponential</u>                             |
|-----------------------------|--------------------------------|--------------------------------------------------------|
| $p = \frac{\pi n \hbar}{L}$ | $E = \frac{(\hbar k_n)^2}{2m}$ | $\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$ |

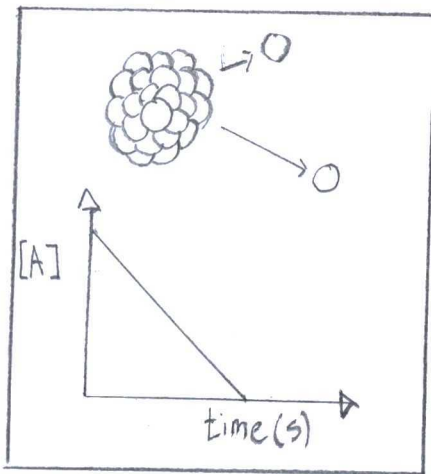
(from book) "Poisson Summation Formula"

$$\sum_{m=-\infty}^{\infty} f(m) = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} d\phi F(\phi) e^{2\pi i n \phi}$$



# Exercise #7:

(pg 118)



"Heavy nucleus having a finite rate of  $\alpha$ -decay"

(from problem) "Effective Potential"

$$U(r) = 2(Z-1)e^2/r$$

Temperature:

$$\frac{3}{2} k_B T = U(r) - E$$

$$T = \frac{2}{3 k_B} [U(r_0) - E]$$

Mean Energy:

$$\bar{E}_m = -\frac{d \ln Q}{d \beta}$$

$$= -\frac{d}{d \beta} \ln \sum_{n=1}^{\infty} e^{-E_n/\beta}$$

$$= -\frac{d}{d \beta} \ln \left( \frac{e^{-\beta E}}{1 - e^{-\beta E}} \right)$$

$$= \frac{\beta e^{\beta E}}{e^{\beta E} - 1}$$

$$= \frac{\beta}{1 - e^{-\beta E}}$$

$$= \frac{k_B T}{1 - e^{-E/k_B T}}$$

$$= \frac{\frac{3}{2} (U(r_0) - E)}{1 - e^{-\frac{3}{2} (U(r_0) - E)/k_B T}} = \frac{3(Z-1)e^2/r_0 - E}{2 \left( 1 - e^{-\frac{3}{2} (2(Z-1)e^2/r_0 - E)/k_B T} \right)}$$

(1850-1900)

Kinetic in ideal  
Energy in gases

$$K.E = \frac{3}{2} k_B T$$

(1867)

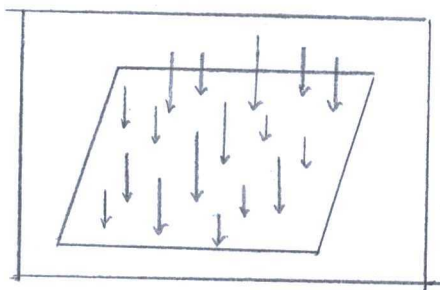
Mean free energy

$$E = -\frac{d \ln Q}{d \beta}$$

Where  $Q$  = canonical ensemble

# Exercise #8

(pg 113)



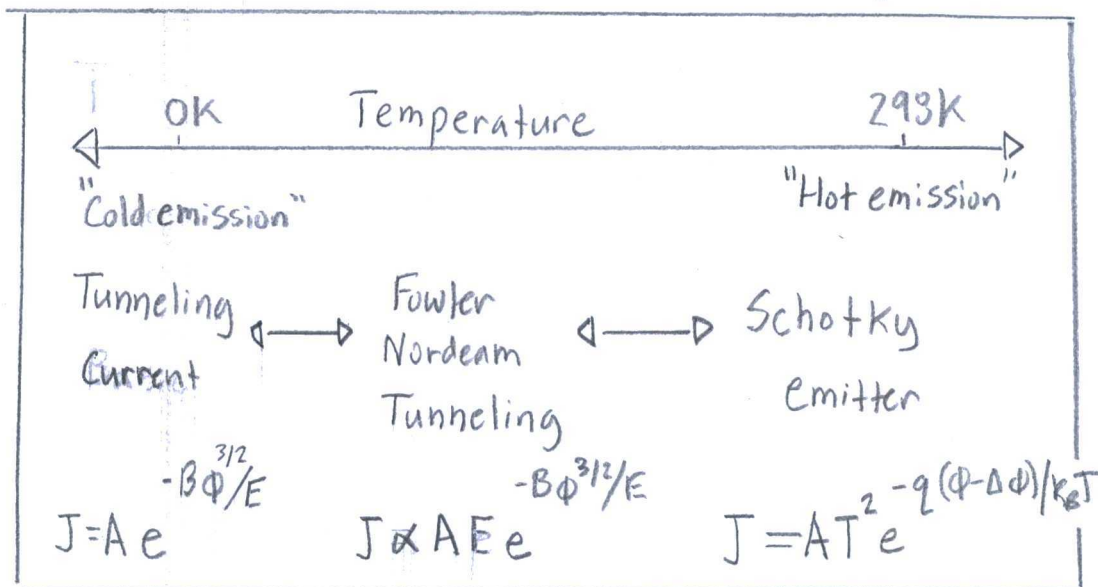
"Uniform electric field... applied to the surface of metal"

Cite: Fermi, Enrico

"Sulla quantizzazione del gas perfetto monoatomico"  
Italy (1926)

Fermi limited the classical equation for energy ( $\frac{3}{2}k_B T$ ) at Zero degrees Kelvin. Other tid bits until 1962, then 1999 in laboratory.

Current from electron tunneling:



...When A, B = Constants specific to metal, Shape, and temp.

$\Phi$  = Work function

E = Energy Fermi gas =  $\frac{3}{2} k_B T$

Exercise in

Edition no. 2

(pg 139)

(1.17) "Euler-Lagrange equation"

$$\forall x, i: \frac{\partial \mathcal{L}}{\partial \phi^i(x)} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi^i(x)} = 0$$

$$\mathcal{L}(\theta, \phi) = \vec{B} \cos \theta + i(1 - \cos \theta) \partial_\tau \phi$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = i \dot{\phi} \sin \theta$$

$$= B_x \cos \theta \cos \phi + B_y \cos \theta \sin \phi - B_z \sin \theta$$

$$\frac{\partial}{\partial \tau} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 0$$

$$= B_x \cos \theta - B_y \cos \phi$$

Block Function:

$$\text{IF } B = B_x + B_y + B_z$$

$$\text{and } n = \sin(\theta) \cos(\theta) \hat{x} + \sin \theta \sin \theta \hat{y} + \cos \theta \hat{z}$$

$$i \partial_\tau n = i \dot{n}$$

$$= \begin{pmatrix} \dot{\theta} \cos \theta \cos \phi - \dot{\phi} \sin \theta \sin \phi \\ \dot{\theta} \cos \theta \sin \phi + \dot{\phi} \sin \theta \cos \phi \\ - \dot{\theta} \sin \theta \end{pmatrix}$$

$$= \begin{pmatrix} i(B_x \sin \phi - B_y \cos \phi) \cos \theta \cos \phi - (B_x \cos \theta \cos \phi + B_y \cos \theta \sin \phi - B_z \sin \theta) \sin \phi \\ i(B_x \sin \phi - B_y \cos \phi) \cos \theta \sin \phi + i(B_x \cos \theta \cos \phi + B_y \cos \theta \sin \phi - B_z \sin \theta) \cos \phi \\ -i(B_x \sin \phi - B_y \cos \phi) \end{pmatrix}$$

$$= \begin{pmatrix} B_y \cos \theta - B_z \sin \theta \sin \phi \\ -B_x \cos \theta + B_z \sin \theta \cos \phi \\ B_x \sin \phi \sin \theta - B_y \sin \theta \cos \phi \end{pmatrix}$$

$$= -\hbar \times \vec{B}$$

$$= B \times n$$

Ex Exercise in  
edition no. 2.  
(pg 145)

(3.6) "Hamiltonian Path Integral"

$$\langle q_f | e^{-iHt/\hbar} | q_i \rangle = \int Dq \, e^{\left( \frac{i}{\hbar} \int_0^t dt' (p \dot{q} - H(p, q)) \right)}$$

$$G(q_f, q_i, t) = \sum_{n=-\infty}^{\infty} \langle q_f | e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} | q_i - 2na \rangle - \langle q_f | e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} | q_i + 2na \rangle$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{-\frac{i}{\hbar} p q_f}}{\sqrt{2\pi\hbar}} \cdot e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} \cdot \frac{e^{-\frac{i}{\hbar} (q_i - 2na)}}{\sqrt{2\pi\hbar}} \cdot dp$$

$$- \int_{-\infty}^{\infty} \frac{e^{-\frac{i}{\hbar} p q_f}}{\sqrt{2\pi\hbar}} \cdot e^{-\frac{i}{\hbar} \frac{\hat{p}^2 t}{2m}} \cdot \frac{e^{-\frac{i}{\hbar} (q_i + 2na)}}{\sqrt{2\pi\hbar}} \cdot dp$$

$$= \sum_{n=-\infty}^{\infty} \left( \frac{1}{2\pi\hbar} \right) \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \left[ \frac{\hat{p}^2 t}{2m} + \frac{(q_f - q_i - 2na)(t)}{t} \right]} \cdot dp$$

$$- \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \left[ \frac{\hat{p}^2 t}{2m} + \frac{(q_f - q_i + 2na)(t)}{t} \right]} \cdot dp$$

$$= \frac{m}{2\pi i \hbar t} \sum_{n=-\infty}^{\infty} e^{i m (q_F - q_i + 2na)^2 / 2 \hbar t} - e^{i m (q_F + q_i - 2na)^2 / 2 \hbar t}$$

Exercise #9:  
(pg 146)

(3.71) "Field Integral"

$$Z = \int D\psi e^{-S[\psi]}, \quad S[\psi] = \int_0^\beta d\tau (\bar{\psi} \partial_\tau \psi + H(\bar{\psi}, \psi))$$

$$\text{where } D\psi = \lim_{N \rightarrow \infty} \prod_{n=1}^N d\psi^n$$

(3.69) "Hamiltonian for a two body interaction"

$$\hat{H}(a^\dagger, a) = \sum_{ij} h_{ij} a_i^\dagger a_j + \sum_{ijkl} V_{ijkl} a_i^\dagger a_j^\dagger a_k a_l$$

(from Problem) "Transformation"

$$\psi = e^{i\phi} \psi, \quad \bar{\psi} = \bar{\psi} e^{-i\phi}$$

$$S[e^{i\phi} \psi] = \int_0^\beta d\tau (\bar{\psi} e^{-i\phi} \partial_\tau \psi e^{i\phi} + H(\bar{\psi} e^{-i\phi}, \psi e^{i\phi}))$$

$$= \int_0^\beta d\tau (\bar{\psi} e^{-i\phi} \partial_\tau \psi e^{i\phi} + \sum_{ij} h_{ij} \bar{\psi} e^{-i\phi} \psi e^{i\phi}$$

$$+ \sum_{ijkl} V_{ijkl} \bar{\psi}_i e^{-i\phi} \bar{\psi}_j e^{-i\phi} \psi_k e^{i\phi} \psi_l e^{i\phi})$$

$$= \int_0^\beta d\tau (\bar{\psi} \partial_\tau \psi + H(\bar{\psi}, \psi))$$



$$= S[\psi]$$

Noether current:

(1.43) "Noethers Theorem"

$$j_\mu = \left( \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^i)} \partial_\nu \phi^i - \mathcal{L} \delta_{\mu\nu} \right) \frac{\partial x_\nu}{\partial w_a} \Big|_{w=0} - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^i)} F_a^i$$

$$\text{where } \mathcal{L} = \bar{\psi} \partial_\tau \psi + H(\bar{\psi}, \psi)$$

$$j_\mu = \bar{\psi} e^{-i\phi} \cdot i \psi e^{i\phi} - i \bar{\psi} e^{-i\phi} \cdot \psi$$

$$= 0$$

Exercise #10:  
(pg 147)

(3.89) "Invariant Transformation"

$$\psi \rightarrow e^{i\phi_a \gamma^5} \cdot \psi, \quad \bar{\psi} \rightarrow \bar{\psi} e^{-i\phi_a \gamma^5}$$

$$\text{where } [\gamma^\mu, \gamma^5] = 0 \text{ for } \mu=0,1$$

The invariant transformation describes a real world observable with equal values in  $2\pi$ -increments, such as interference by a wave.

A function for the wave is

$$\psi(x) = e^{i(kx - \omega t)}$$

(3.86) "One-dimensional electron gas"

$$S_0[\psi] = \int d^2x \psi^\dagger (\sigma_0 \partial_{x_0} - i \sigma_3 \partial_{x_1}) \psi$$

$$= \int d^2x \psi (\sigma_1 \partial_{x_1} - i \sigma_2 \partial_{x_1}) \psi$$

(from problem) "Transformation"

$$\psi \rightarrow e^{i\phi_a \sigma_3} \psi, \quad \bar{\psi} \rightarrow \bar{\psi} e^{-i\phi_a \sigma_3}$$

$$S_0[e^{i\phi_a \sigma_3} \psi] = \int d^2x \bar{\psi} e^{-i\phi_a \sigma_3} (\sigma_1 \partial_{x_0} - \sigma_2 \partial_{x_1}) \psi e^{i\phi_a \sigma_3}$$

$$= \int d^2x \bar{\psi} (\sigma_1 \partial_{x_0} - \sigma_2 \partial_{x_1}) \psi$$

$$= S_0[\psi]$$

$$S_0[\bar{\psi} e^{i\phi_a \sigma_3}] = \int d^2x \bar{\psi} e^{-i\phi_a \sigma_3} (\sigma_1 \partial_{x_0} - \sigma_2 \partial_{x_1}) \psi e^{i\phi_a \sigma_3}$$

$$= S_0[\bar{\psi}] \quad \dots \text{with } \phi_a = 2\pi$$

$$S_0[\psi e^{i\phi_a \sigma_3}] = \int d^2x \bar{\psi} (\sigma_1 \partial_{x_0} - \sigma_2 \partial_{x_1}) \psi e^{i\phi_a \sigma_3}$$

$$= S_0[\psi] \quad \dots \text{with } \phi_a = 2\pi$$

Exercise #11:  
(pg 149)

(3.93) "Anticommutator identity"

$$[e^{i\hat{\sigma}\theta(x)}, e^{i\hat{\phi}(x)}] = [e^{i\hat{\sigma}\theta(x')}, e^{i\hat{\phi}(x')}]$$

$$\propto e^{i\pi\theta(x'-x)} + e^{i\pi\theta(x-x')} = 0$$



(3.92) "String operator"

$$[\hat{\phi}(x), \Theta(y)] = -i\pi\Theta(y-x)$$

$$[e^{iA} e^{iB}, e^{iA'} e^{iB'}] =$$

$$= e^{iA} e^{iB} e^{iA'} e^{iB'} + e^{iA'} e^{iB'} e^{iA} e^{iB} - [\hat{\phi}(x), \hat{\theta}(x)] [\hat{\phi}(x'), \hat{\theta}(x')] = e^{i\pi\Theta(x'-x)} + e^{i\pi\Theta(x-x')} = e$$

where  $[\hat{\phi}(x), \hat{\theta}(x)] = -i\pi\Theta(x-x')$

Canonical Anticommutation relation:

$$[c_i, c_j^\dagger] = \frac{1}{\sqrt{2}} e^{i s \theta_i} e^{i \phi_i} \frac{1}{\sqrt{2}} e^{-i \phi_j} e^{-i s \theta_j} + \frac{1}{\sqrt{2}} e^{-i \phi_i} e^{-i s \theta_i} \frac{1}{\sqrt{2}} e^{i s \theta_j} e^{i \phi_j} = \frac{1}{2} (e^{-[s \theta_i, \phi_j]} + e^{-[s \theta_j, \phi_i]}) = e^{i\pi\theta} = 1 \quad \text{when } [s \theta_i, \phi_i] = \delta_{ij} = 1$$

Commutator

identities

$$e^{\hat{A}} e^{\hat{B}} e^{-\hat{A}} = e^{[\hat{A}, \hat{B}]} e^{\hat{B}}$$

$$e^{\hat{A}} e^{\hat{B}} e^{\hat{A}} = e^{\hat{A}} e^{\hat{B}} e^{-\hat{A}}$$

Exercise #12: (pg 151) "Lagrangian from acting integral"  
(pg 151)

$$S_M[\theta] = \frac{1}{2\pi} \int d\tau dx ((\partial_\tau \theta)^2 - (\partial_x \theta)^2)$$

The book wants a "quick" alternative.

$$(\partial_x \phi + i \partial_\tau \theta)^2 = 0, \text{ so } (\partial_\tau \theta)^2 = -(\partial_x \phi)^2 - 2i \partial_\tau \partial_x \phi$$

$$\begin{aligned} S_M[\theta] &= \frac{1}{2\pi} \int d\tau dx (-(\partial_x \phi)^2 - 2i \partial_\tau \partial_x \phi - (\partial_x \theta)^2) \\ &= -S[\theta] \end{aligned}$$

$$S[\theta] = \frac{1}{2\pi} \int d\tau dx ((\partial_x \phi)^2 + 2i \partial_\tau \partial_x \phi + (\partial_x \theta)^2)$$

Site: СТРАТОНОВИЧ, Р.Л.

"ОБ ОДОМ МЕТОДЕ ВЫЧИСЛЕНИЯ  
КВАНТОВИХ ФУНКЦИИ  
РАСПРЕДЕЛЕНИЯ"

Доклады Академии Наук, СССР  
(1957)

J. Hubbard (1959) describes СТРАТОНОВИЧ  
as a solution to Bose-Einstein condensates

by chemical potential. The syntax matches, but without bounds.

Exercise #13: (1.43) "Noether Current"  
(pg 152)

$$j^{\mu\nu} = \left( \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)} \partial_\nu \phi^i - \mathcal{L} \delta^\mu_\nu \right) \frac{\partial x^\nu}{\partial w_a} \Big|_{w=0} - \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)} F_a^i(\phi)$$

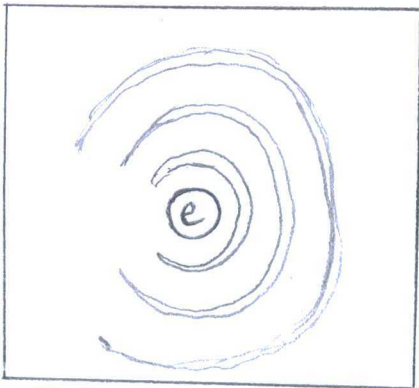
$$= \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)} \eta^\mu$$

$$j^0_0 = \rho \quad j^0_1 = -i \quad j^1_0 = \rho_+ - \rho_- \quad (\text{pg 151})$$

$$= \frac{\partial_\nu \theta}{\pi} \quad = \frac{-i \partial_x \phi}{\pi} \quad = \frac{1}{2\pi} (\partial_x \theta + \partial_x \phi) - \frac{1}{2\pi} (\partial_x \theta - \partial_x \phi)$$

$$= \frac{\partial_x \phi}{\pi}$$

Exercise #14:  
(pg 163)



"Friedel oscillation"

(from problem) "Density-Density  
Correlation function"

$$\Pi(x, t) = \langle \hat{\rho}(x, t) \rho(0, 0) \rangle - \langle \hat{\rho}(x, t) \rangle \langle \hat{\rho}(0, 0) \rangle$$

where  $\hat{\rho} = \text{"Groundstate"}$   
 $= a^\dagger a$

$$a(x) = e^{ik_F x} a_+(x) + e^{-ik_F x} a_-(x)$$

"Left - and - right wave"

(from problem) "Von neumann equation"

$$\dot{a}_{sq} = i [\hat{H}, a_{sq}]$$

$$\hat{H} = \sum_{q,s} V_F (p_F + s q) a_{sq}^\dagger a_{sq}$$

Time dependent operator:

$$\dot{a}_{sq} = \frac{da}{dt}$$

$$= i [\hat{H}, a_{sq}]$$

$$= i [\hat{H} a_{sq} - a_{sq} \hat{H}]$$

$$\frac{da}{a_{sq}} = -i V_F (p_F + s q) dt$$

$$a_{sq} = e^{-i V_F (p_F + s q) t}$$

The static response decays

with a maximum  $|x|^{-1}$  where

the expectation (average) is

$1/x^2$  from the correlation function.

$$\Pi(x,t) = \frac{1}{4\pi} \left( \frac{1}{(x-v_F t)^2} + \frac{1}{(x+v_F t)^2} + \frac{2\cos(2p_F x)}{x^2 - (v_F t)^2} \right)$$

### Problem 3.8.1 - Quantum Harmonic Oscillator:

a) (3.71) "Field integral"

$$Z = \int D\psi e^{-S[\psi]}, \quad S[\psi] = \int_0^\beta d\tau (\bar{\psi} 2\pi \dot{\psi} + H(\bar{\psi}, \psi))$$

(pg 102) "Quantum Mechanical amplitude"

$$Z = \int dq \langle q | e^{-i\hat{H}/\hbar} | q \rangle \Big|_{\substack{\hbar=1/\beta \\ t=-iL}}$$

(from problem) "Hamiltonian"

$$\hat{H} = \hat{p}^2/2m + m\omega^2 \hat{q}^2/2$$

$$\text{So, } S[q] = \int_0^t dt \hat{H}$$

$$= \int_0^t dt \left( \frac{\hat{p}^2}{2m} + \frac{m\omega^2 \hat{q}^2}{2} \right)$$

$$= \int_0^t dt' \frac{m}{2} (\dot{q}^2 - \omega^2 q^2)$$

$$= \int_0^t dt \frac{m}{2} \left( \left( \frac{d}{dt} (A \sin(\omega t) + B \cos(\omega t)) \right)^2 - \omega^2 (A \sin(\omega t) + B \cos(\omega t))^2 \right)$$



$$= \frac{m\omega}{4} \left[ (A^2 - B^2) \sin(2\omega t) - 4AB \sin^2(\omega t) \right]$$

$$= \frac{m\omega^2}{2} \left[ (A^2 - B^2) \frac{\sin(2\omega t)}{2\omega} + \frac{2AB \cos(2\omega t) - 1}{2\omega} \right]$$

Identity:

$$\sin^2(\omega t) = \frac{1 - \cos(2\omega t)}{2}$$

$$= \frac{m\omega}{2} \left[ (q_i^2 + q_f^2) \cot(\omega t) - \frac{2q_i q_f}{\sin(\omega t)} \right]$$

as for  $A = q_i$  and  $B = q_f$

$$\begin{aligned} \langle q_f | e^{-iHt/\hbar} | q_i \rangle &= \int_{q[0]=q_i}^{q[t]=q_f} Dq e^{\frac{i}{\hbar} \int_0^t L(q, \dot{q}) dt} \\ &= \int_{q[0]=q_i}^{q[t]=q_f} Dq e^{\frac{i}{\hbar} \frac{m\omega}{2} \left[ (q_i^2 + q_f^2) \cot(\omega t) - \frac{2q_i q_f}{\sin(\omega t)} \right]} \end{aligned}$$

$$= \left( \frac{m\omega}{2\pi i \hbar \sin(\omega t)} \right) e^{\frac{i}{\hbar} \frac{m\omega}{2} \left[ (q_f^2 + q_i^2) \cot(\omega t) - \frac{2q_i q_f}{\sin(\omega t)} \right]}$$

The singularities at  $t = n\pi/\omega$  for  $n=1, 2, \dots$  associate to uniform Spring models. The oscillator is in ground state.



b) Wavepacket :  $\psi(q, t=0) = (2\pi a)^{-1/4} \exp[-q^2/4a]$

$$\psi(q) = J(t) \int_{-\infty}^{\infty} dq' \frac{1}{(2\pi a)^{1/4}} e^{\frac{i}{\hbar} \frac{m\omega}{2} [(q_F^2 + q_i^2) \cos(\omega t) - \frac{2q_i q_F}{\sin(\omega t)}]}$$

$$= J(t) \int_{-\infty}^{\infty} dq' \frac{1}{(2\pi a)^{1/4}} e^{\beta^2/2\alpha} e^{\left(\frac{i}{2\hbar} m\omega q^2 \cos(\omega t)\right)}$$

where  $\beta = i m \omega q / \hbar \sin(\omega t)$

$$\alpha = \frac{1}{2} a + i m \omega \cos(\omega t) / \hbar$$

$$= J(t) \frac{1}{(2\pi a)^{1/4}} \sqrt{\frac{2\pi}{\alpha}} e^{\beta^2/2\alpha} e^{\left(\frac{i}{2\hbar} m\omega q^2 \cos(\omega t)\right)}$$

where  $\beta^2/2\alpha = -(1 + i k \cos(\omega t) q^2 / 4a(t))$

$$= \frac{1}{(2\pi a(t))^{1/4}} e^{-q^2/4a(t)} e^{i\phi(q,t)}$$

Width:

$$a(t) = a(\cos^2(\omega t) + K^2 \sin^2(\omega t))$$

where  $K = 2am\omega/\hbar$

c) Initial wave packet:

$$\psi(a, t=0) = (2\pi a)^{-1/4} e^{\frac{i}{\hbar} m v q - \frac{1}{4a} q^2}$$

Wavepacket at t:

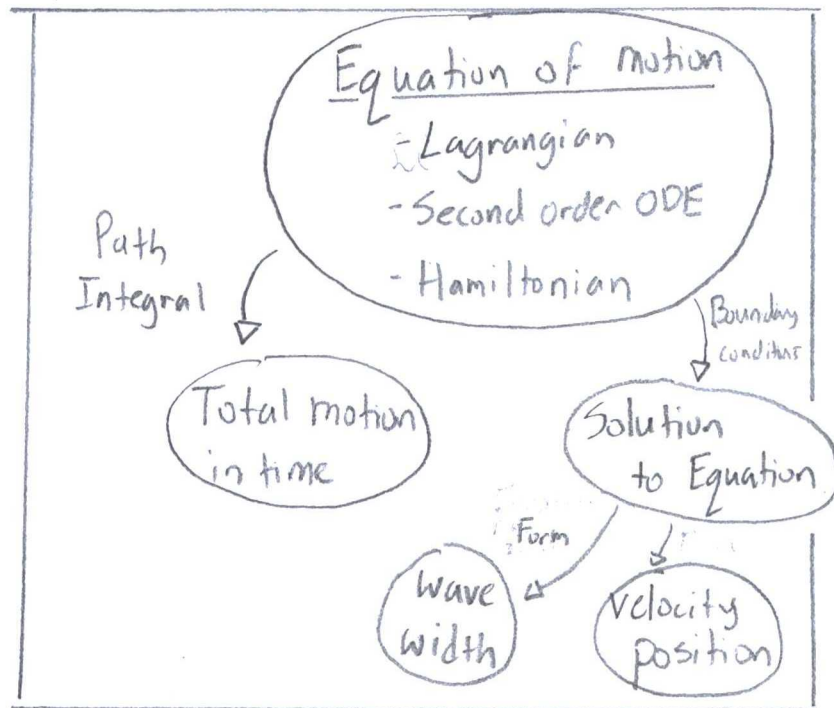
... same as  $b$ , except  $\beta = \frac{i}{\hbar} \frac{m\omega}{\sin(\omega t)}$

$$x(q - \frac{v}{\omega} \sin(\omega t))$$

Position:  $q_0(t) = (v/\omega) \sin(\omega t)$

Velocity:  $v(t) = v_0 \cos(\omega t)$

### Problem 3.8.2: Density Matrix:



$$\rho(q, q') = \langle q_f | e^{-i\hat{H}t/\hbar} | q_i \rangle$$

$$= \langle q | e^{-\beta H} | q' \rangle$$

$$= \langle q | e^{-(i/\hbar) \beta H(\hbar/i)} | q' \rangle$$

$$= \left( \frac{m\omega}{2\pi i \hbar \sin(i\beta \hbar \omega)} \right) e^{\frac{i}{2\hbar} m\omega((q_i^2 + q_f^2) \cot(\beta \hbar \omega) - \frac{2q_i q_f}{\sin(i\beta \hbar \omega)})}$$

### (3.76) "Partition function"

$$\begin{aligned}
 Z &= \int D\phi e^{-S[\phi]} \\
 &= \int D\phi e^{-\int d\phi_n d\phi_n S[\phi]} \\
 &= \int D\phi e^{-M \int d\phi d\phi S[\phi]} \\
 &= \int D\phi e^{-\prod_{\omega} \begin{pmatrix} -i\omega + \xi & \mu_0 B / 2\sigma_{\text{eff}} \\ \mu_0 B / 2\sigma_{\text{eff}} & -i\omega + \xi \end{pmatrix} \int_0^B \int_0^B d\phi d\phi \phi^* \phi} \\
 &= \int D\phi e^{-\beta^2 [(-i\omega + \xi)^2 - (\frac{\mu_0 B}{2})^2]} \\
 &= \int D\phi e^{-\prod_n (\beta^2 (-i\omega_n + \xi_n)^2 - (\frac{\mu_0 B}{2})^2)}
 \end{aligned}$$

↓ similar to  
 equation (3.76)  
 - unsure how to  
 plot, or table-  
 table

### Free energy:

### (3.77) "Free Energy"

$$\begin{aligned}
 F &= -T \ln Z \\
 &= -T \ln \left[ \prod_n \beta^2 (-i\omega_n + \xi)^2 - \left( \frac{\mu_0 B}{2} \right)^2 \right] \\
 &= -T \sum \ln \left[ \beta^2 (-i\omega_n + \xi)^2 - \left( \frac{\mu_0 B}{2} \right)^2 \right]
 \end{aligned}$$

b) Magnetic Susceptibility:

(from problem) "Total Energy change"

$$\Delta E \sim -\mu_0^2 B^2 \rho(u)$$

(from problem) "Magnetic Susceptibility"

$$\chi \sim -\partial_B^2 \Delta E$$

$$= -\mu_0^2 \left. \frac{\partial^2}{\partial B^2} \right|_{B=0} \Delta E$$

$$= -\mu_0^2 \left. \frac{\partial^2}{\partial B^2} \left[ -T \sum \ln(\beta^2 [(-i\omega_n + \xi)^2 - \frac{1}{4}(\mu_0 B)^2]) \right] \right|_{B=0}$$

$$= -\frac{1}{2} \mu_0^2 T \sum \frac{1}{(-i\omega_n + \xi)^2}$$

(Pg 142) "Isolated Singularities"

$$S = \sum h(\omega_n)$$

$$= -\frac{1}{2} \mu_0^2 T \sum_{\alpha} \frac{1}{(-i\omega_n + \xi)^2}$$

... Where  $h(\omega_n) = n_F(\epsilon_n)$

$$= -\frac{1}{2} \mu_0^2 \sum_{\alpha} n_F(\epsilon)$$

$$\lim_{T \rightarrow 0} \chi = -\frac{\mu_0^2}{2}$$

Problem 3.8.10 - Boson-Fermion duality:

Matsubara :

Re(Frequency)  $\leftrightarrow$  Im(Frequency)

Fourier :

Re(Frequency)  $\leftrightarrow$  Re(Time)

Laplace :

Re(Time)  $\leftrightarrow$  Im(Time)

Hankel :

Re(position)  $\leftrightarrow$  Re(scaled position)

Wavelet :

Re(time)  $\leftrightarrow$  Re(scaled time)

(3.71) "Partition function"

$$\begin{aligned} Z &= \int D\psi e^{-S[\psi]} \\ &= \int D\psi e^{-\int dx d\tau \psi^\dagger (\partial_{x_0} \pm i\partial_x) \psi} \\ &= \int D\psi e^{-\int dx d\tau \psi^\dagger G(0,0,t)^{-1} \psi} \quad \dots (3.29) \\ &= \int D\psi e \end{aligned}$$

$$G_{\pm}(x, \tau) = (\partial_{\tau} \pm \partial_x)^{-1}$$

(3.95) "Correlation function"

$$\begin{aligned} C(x, \tau) &\equiv \langle (\psi_+^\dagger \psi_-)(x, \tau) (\psi_-^\dagger \psi_+)(0, 0) \rangle_{\psi} \\ &\equiv G_0(x, \tau) \circ G_0(x, \tau) \end{aligned}$$

"Transforms

b) (from problem) "Action in a Bosonic field"

$$\begin{aligned} S[\theta] &= \frac{1}{2c} \int dx d\tau [(\partial_{\tau} \theta)^2 + (\partial_x \theta)^2] \\ &= \frac{L}{2cT} \sum_{x, \tau} [q^2 + \omega^2] \\ &= G(x, \tau) \end{aligned}$$

Correlation function :



$$K(x, \tau) = \langle \theta(x, \tau) \theta(0, 0) - \theta(0, 0) \theta(0, 0) \rangle$$

$$= G(x, \tau)$$

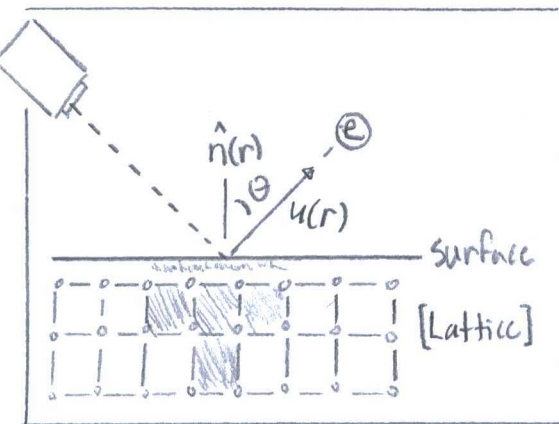
C) (from problem) "Correlation Function"

$$C(x, \tau) = T^2 \langle e^{2i\theta(x, \tau)}, e^{-2i\theta(0, 0)} \rangle$$

$$= T^2 \langle e^{2i(\theta(x, \tau) - \theta(0, 0))} \rangle$$

$$= T^2 G(x, \tau)$$

### Problem 3.8.11 - Electron Phonon Coupling



"Electron-phonon coupling"

(from problem) "Electron-phonon Hamiltonian"

$$\hat{H}_{el-ph} = \gamma \int d^3r \hat{n}(r) \nabla \cdot u(r) = \gamma \sum_{q,j} \frac{i q_j}{(2m\omega_q)^{1/2}} \hat{n}_q (a_q + a_{-q}^\dagger)$$

"Angle between normal"

$$\text{where } u(r) = e_j (a_{q,r} + a_{-q,r}^\dagger) / (2m\omega_q)^{1/2}$$

(from problem) "Phonon Hamiltonian"

$$\hat{H}_{ph} = \sum_{q,j} \omega_q a_{q,j}^\dagger a_{q,j}$$



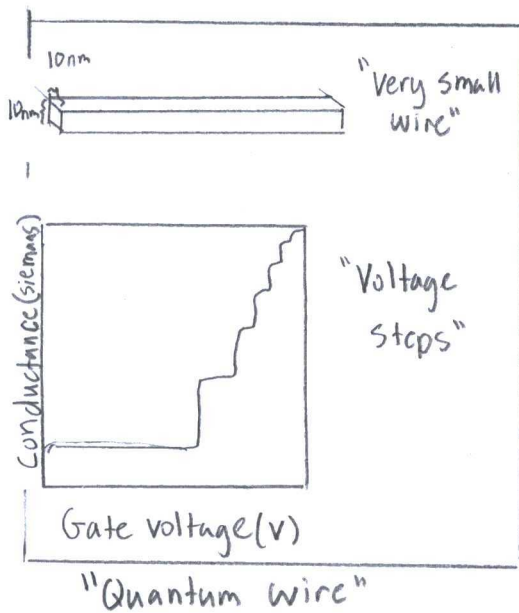
a) (3.71) "Partition Function"

$$\begin{aligned}
 Z &= \int D\phi e^{-S[\phi]} \\
 &= \int D\phi e^{-\left(S[\phi]_{el-ph} + S[\phi]_{ph} + S[cl]\right)} \\
 &= \int D\phi e^{-\int d\phi \gamma \frac{i q_2}{(2m q)^{1/2}} \hat{n}_q (a_q + a_q^\dagger) + \omega a_q^\dagger a_q + \frac{\hat{p}^2}{2m}} \\
 &= \int D\phi e^{-\int d\phi \gamma \frac{i q}{(2m q)^{1/2}} p_q (\phi_{q_i} + \bar{\phi}_{q_i}) + \omega a_q^\dagger a_q + \frac{\hat{p}^2}{2m}} \\
 &= \int D\phi e
 \end{aligned}$$

b)  $S_{eff}[\phi] = S_{el}[\phi] + S_{ph}[\phi] + S_{el-ph}[\phi]$

$$= \frac{\hat{p}^2}{2m} + \hbar \omega - \frac{\gamma^2}{2m\omega} \sum \frac{q^2}{L} \hat{n}_p p p$$

## Problem 3.8.12 - Disordered Quantum Wires:



(3.86) "Action Potential in fermionic gas."

$$\begin{aligned}
 S_0[\psi] &= \int d^2x \psi^\dagger (\sigma_0 \partial_{x_0} - i \sigma_3 \partial_{x_1}) \psi \\
 &= \int d^2x \bar{\psi} (\sigma_1 \partial_{x_0} - \sigma_2 \partial_{x_1}) \psi
 \end{aligned}$$

(3.90) "Action with second quantization"

$$S_{int}[\psi] = \frac{1}{2} \int_s \int dxd\tau (g_s \hat{p}_s \hat{p}_s + g_4 \hat{p}_s \hat{p}_s)$$

(from problem) "Action in impurity"

$$S_{imp}[\psi] = \int d\tau (V_t \psi_+^\dagger \psi_+ + V_l \psi_+^\dagger \psi_- + V \psi_+^\dagger \psi_- + \bar{V} \psi_-^\dagger \psi_-)$$

a) If  $\psi_+ = e^{-i v_t \theta(x)/v_F}$ ,  $S_{impurity}[\psi]$  is

without a  $V_t \psi_+^\dagger \psi_+$  term. The forward scattering term is not present in a "one-dimensional" wire.

b) (pg 151) "Lagrangian form of action"

$$S[\theta] = \frac{1}{2\pi} \int d\tau dx ((\partial_\tau \theta)^2 + (\partial_x \theta)^2)$$

$$S_{Tot}[\theta] = \frac{1}{2\pi} \int dx d\tau (v (\partial_x \theta)^2 + (\partial_\tau \theta)^2) + S_{imp}[\psi]$$

c) (from problem) "Action Integral"

$$e^{-S[\theta]} = e^{-S_{imp}[\theta]} \cdot \int D\theta dk e^{-\int dx d\tau (\frac{1}{2\pi g} (v (\partial_x \theta)^2 + \bar{v} (\partial_\tau \theta)^2) - i k (\hat{\theta} - \theta) \delta(x))}$$

$$= e^{-S_{imp}[\theta]} \cdot \int Dk e^{i \sum k \theta (-\frac{\pi g \Gamma}{2L} \sum (v q^2 + \bar{v} \omega^2) |k_n|^2)}$$